E-COMMERCE RETURN RATE INVESTIGATION REPORT

OBJECTIVE

The objective of this project is to analyze e-commerce return data and identify the key factors influencing product returns. The study aims to understand customer return behavior and provide business recommendations to reduce return rates and improve operational efficiency.

PROBLEM STATEMENT

Product returns are a major challenge in e-commerce businesses as they increase operational costs and reduce profitability. This project investigates return patterns, customer behavior, product characteristics, and delivery performance to identify the major causes of returns and suggest corrective actions.

DATASET DESCRIPTION

The dataset contains information about e-commerce orders, product details, customer ratings, delivery information, discounts, and return status.

Dataset Summary

- Total Orders: 5000
- Total Returned Products: 2500
- Return Percentage: 50.0%
- Average Delivery Duration: 3.35 Days
- Average Product Price: 145.73
- Average Customer Rating: 3.96

The dataset consists of 22 attributes related to customer purchases and return behavior.

STATISTICAL METHODS USED

The following statistical techniques were applied:

1. Mean (Average)
2. Median
3. Standard Deviation
4. Quartile Analysis
5. Correlation Analysis
6. Outlier Detection using IQR Method
7. Customer Segmentation
8. Hypothesis Testing (T-Test)
9. Data Visualization Techniques

ANALYSIS AND FINDINGS

Finding 1: Return Rate Analysis

Out of 5000 orders, 2500 products were returned, resulting in a return rate of 50%.

Finding 2: Delivery Performance

The average delivery duration is 3.35 days. Delivery delays show a noticeable impact on customer return behavior.

Finding 3: Product Pricing

The average product price is 145.73. Product price variations influence customer purchasing and return decisions.

Finding 4: Customer Satisfaction

The average customer rating is 3.96 out of 5. Lower customer ratings are associated with a higher probability of returns.

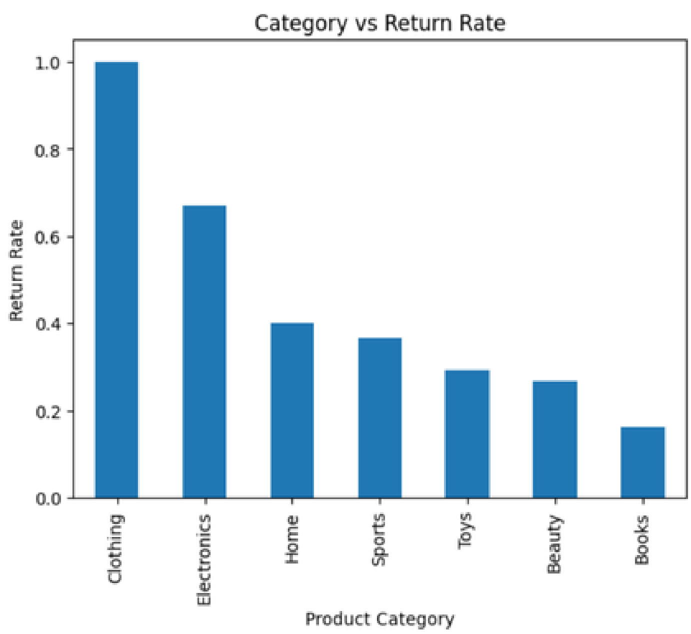
Finding 5: Category-wise Analysis

Certain product categories exhibit higher return rates compared to others, indicating category-specific risks.

VISUALIZATIONS

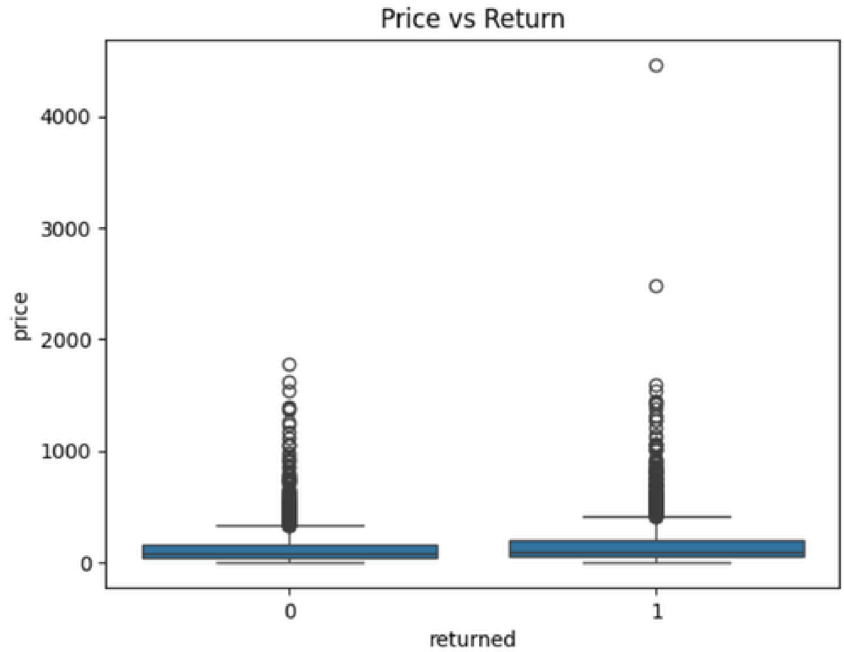
The following visualizations were created during the analysis:

Figure 1: Category vs Return Rate



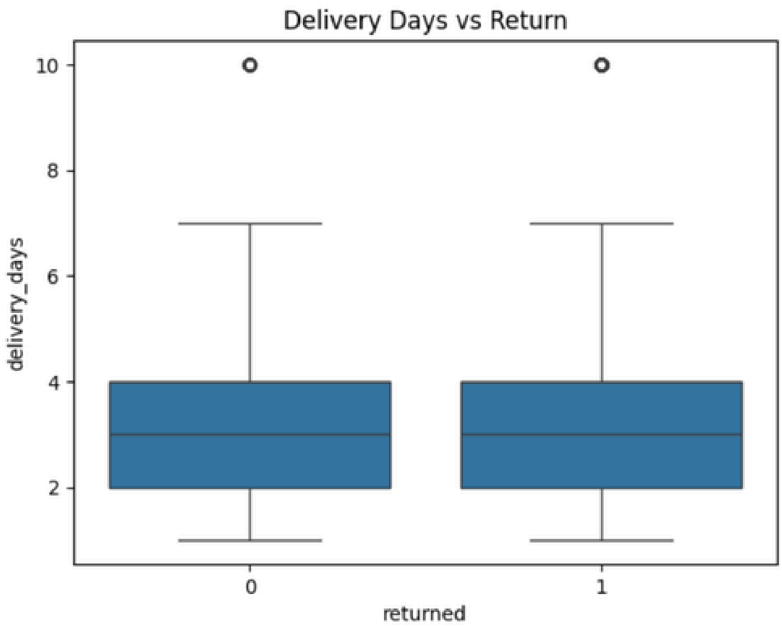
Description: This chart shows the return rate across different product categories.

Figure 2: Price vs Return



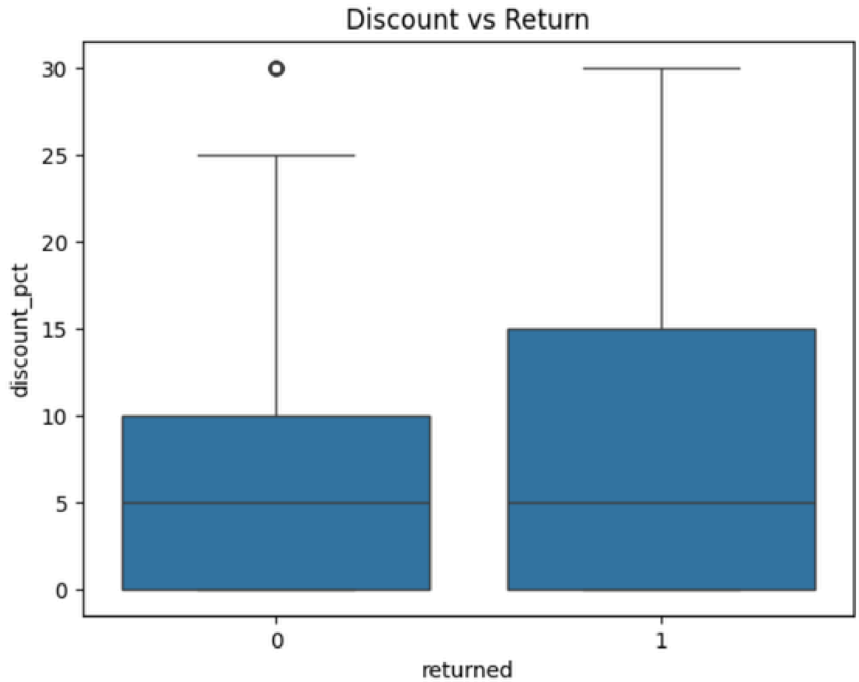
Description: This boxplot illustrates the relationship between product price and return behavior.

Figure 3: Delivery Days vs Return



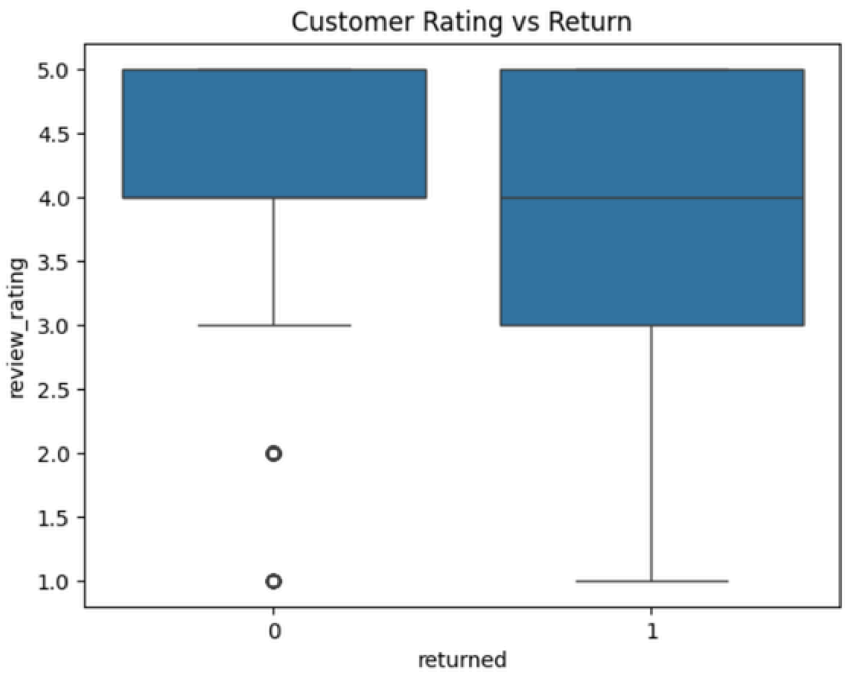
Description: This graph shows how delivery duration affects product returns.

Figure 4: Discount vs Return



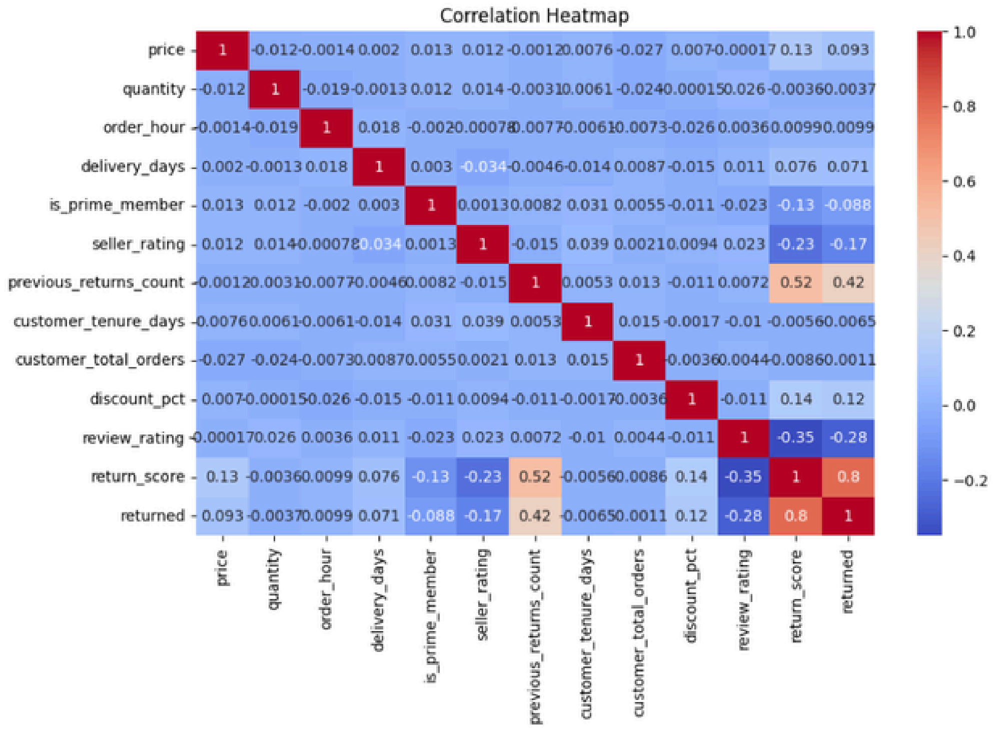
Description: This chart analyzes the impact of discounts on returns.

Figure 5: Customer Rating vs Return



Description: This graph shows the relationship between customer ratings and returns.

Figure 6: Correlation Heatmap



Description: The heatmap displays correlations among numerical variables in the dataset.

BUSINESS RECOMMENDATIONS

1. Improve delivery efficiency and reduce delivery delays.
2. Monitor high-return product categories closely.
3. Strengthen product quality checks before shipping.
4. Review discount strategies for frequently returned products.
5. Improve product descriptions and customer information.
6. Provide proactive customer support for return-prone customers.
7. Analyze customer feedback regularly to identify recurring issues.

FUTURE SCOPE

1. Develop a Machine Learning model to predict product returns.
2. Implement real-time return risk monitoring systems.
3. Improve customer segmentation using advanced analytics.
4. Build automated recommendation systems for return reduction.
5. Use predictive analytics for inventory and logistics planning.

CONCLUSION

The analysis successfully identified key factors affecting e-commerce product returns. Delivery performance, customer ratings, pricing, and product categories play significant roles in return behavior. By implementing the recommended business strategies, organizations can reduce return rates, improve customer satisfaction, and increase operational efficiency.